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Examination	University	Institute	Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2027	6.25
Intermediate	CBSE	Sanskar Public School	2023	Core: 132 Total: 168
Matriculation	CBSE	RKVM	2021	88.40%
				71.50%

SCHOLASTIC ACHIEVEMENTS

- Secured **99.3 percentile** in Joint Entrance Exam (JEE) - Mains from among **1.1+ Million candidates**. [2023]
- Among **top 3 %** in JEE-Advanced 2023 amongst a competitive field of more than **1 lakh 60 thousands** candidates. [2023]

PROFESSIONAL EXPERIENCE

AI/ML & Robotics Intern | Bajaj Auto Ltd. | *Industrial Robotic Pick-and-Place Automation* [May '26 - Present]

- Built an end-to-end robotic vision pipeline integrating **BlenderProc** synthetic data generation, **YOLO (Detectron2)** object detection, **GDRNPP** 6D pose estimation, and a custom **neural network** for automated post-placement verification.
- Developed custom tooling for **camera calibration**, **hand-eye calibration**, **intrinsic/extrinsic parameter estimation**, and **object-to-camera transformation** computation, enabling accurate coordinate-frame alignment for robot manipulation.
- Implemented autonomous grasp planning using **ROS 2 Humble** and **MoveIt 2**, designing custom **state machines** with failure recovery, **grasp-success prediction**, and precise multi-object pick-and-place into constrained cavity geometries.
- Modernized and refactored the **GDRNPP** 6D pose estimation framework for **Python 3.14** compatibility by updating dependencies, resolving legacy issues, and improving maintainability for seamless integration into the perception stack.

Software Engineering Intern | Simphy | *LLM Infrastructure & RAG Systems* [Aug '25 - Sep '25]

- Developed a **custom RAG pipeline** over Simphy API documentation to retrieve relevant context for script generation.
- Built and integrated **LangChain**-based workflows, improving orchestration and interaction with multiple **LLM** providers.
- Evaluated and benchmarked multiple **Vertex AI** foundation models to compare inference quality, latency, and suitability.

CogniCad | Founder & CEO | *AI-Driven CAD Automation Tool* [Feb '26 - Present]

- Secured **INR 25L R&D grant** through IIT Bombay Invent Program in collaboration with Groww.
- Exploring a natural language-to-parametric CAD pipeline generating OpenSCAD scripts via Claude API (Python backend).
- Exploring an agentic execution layer using on top of FreeCAD via sockets, enabling iterative tool use for geometry generation.

KEY PROJECTS

Telemeter | Founder | *Flutter, Python, ROS 2, MQTT, Time-Series Analytics* [Oct '24 - Present]

- Built a telemetry platform using **Flutter** and **Python** for real-time visualization, logging, and analysis of telemetry streams.
- Developed data ingestion pipelines supporting serial and network interfaces with preprocessing, storage, and live dashboard.
- Re-writting Telemeter as a modular telemetry platform with scalable backend services and plugins/scripting type architecture.

Scribbler | Open Source Contributor [Mar '24 - May'25]

- Implemented the **Monokai** theme in dark mode for a more user-friendly interface and better visuals in dark mode.
- Added support for **top-level await** in the JavaScript notebook, enabling it to function as if running in a module.
- Collaborated closely** with the founder on design improvements, feature development, and overall project direction.

POSITION OF RESPONSIBILITY

Institute Technical Secretary | *Tinkerers' Laboratory* [Apr '25 - Mar '26]

- Pitched** and **coordinated** with alumni batches (1966, 1975, 1976) to secure **INR 3 Cr in funding**, for expansion.
- Led** the end-to-end effort to relocate Tinkerers' Lab to a $5\times$ larger space (**5,000 sq. ft.**), overseeing planning and transition.
- Launched Paid Project Internships with **Y Point Combinator projects**, secured internships for 8 students out of 50.
- Spearheading** team of **55** volunteers, **4** salaried employees to run a **24/7** active tinkering lab for **25K+ students** of IITB.
- Mentored and guided **10+** student projects and supported 2 teams in securing **INR 15 lakh** total funding from IIT Bombay.

TECHNICAL SKILLS

Programming	Python, C/C++, Java, MATLAB, JavaScript/TypeScript, Lua, Rust, Dart, Flutter
Frameworks/ Libraries	MERN Stack(MongoDB, Express, React, Node.js), Django, Flutter, ROS, OpenCV, TensorFlow/PyTorch
Softwares/ OS	SSolidworks, Fusion360, OpensCAD, FreeCAD, Docker, Git/GitHub, EasyEDA, Arduino IDE, Arch Linux, Kubernetes